

Essentials of Data Science

Laboratory - 2304102L - 2026

Search course

ctrl + k

3.2.7. Student Data Analysis and Operations

4. Practical 4

4.1. Practice Lab Assignment

4.1.1. Pandas - series creation and manipu...

4.1.2. Dictionary to dataframe

4.1.3. Student Information

4.2. Lab Assignment

4.2.1. Month with the Highest Total Sales

4.2.2. Best Selling Product

4.2.3. City that Sold the Most Products

4.2.4. Most Frequently Sold Product Pairs

4.2.5. Titanic Dataset Analysis and Data CL...

4.2.6. Titanic Dataset Analysis and Data CL...

4.2.7. Titanic Dataset Analysis and Data CL...

4.2.8. Titanic Dataset Analysis and Data CL...

5. Practical 5

5.1. Practice Lab Assignment

5.1.1. Stacked Plot

5.2. Lab Assignment

5.1.1. Stacked Plot02:12

Create a stacked area plot to visualize the temperature variations for three different cities (City A, City B, and City C) across the months of the year. The temperature data is provided for each city in the editor.

Your task is to:

- Create a stacked area plot using the data.
- Label the x-axis as "Month", the y-axis as "Temperature", and provide the title "Temperature Variation" for the plot.
- Display the plot showing the temperature variation for each city throughout the months of the year.

Sample Test Cases

stackedpl...

1import matplotlib.pyplot as plt
2import pandas as pd
3
4# Data for Months and Temperature for three cities
5data = {
6Month: ['January', 'February', 'March', 'April', 'May', 'June',
7July', 'August', 'September', 'October', 'November', 'December'],
8City_A_Temperature: [5, 7, 10, 13, 17, 20, 22, 21, 18, 12, 8, 6],
9City_B_Temperature: [2, 3, 5, 6, 10, 14, 16, 17, 12, 9, 5, 3],
10City_C_Temperature: [3, 4, 6, 8, 9, 12, 15, 14, 10, 7, 4, 2]
11
12# The code from your image
13plt.stackplot(data['Month'], data['City_A_Temperature'],
14data['City_B_Temperature'], data['City_C_Temperature'])
15plt.xlabel('Month')
16plt.ylabel('Temperature')
17plt.title('Temperature Variation')
18plt.show()

Average time0.126 s126.00 msMaximum time0.126 s126.00 ms1 out of 1 shown test case(s) passed

Test case 1126 msDebug

Expected outputActual output

Temperature VariationTemperature Variation

TerminalTest cases

Course

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4.2.4. Most Frequently Sold Product Pairs

4.2.5. Titanic Dataset Analysis and Data CL...

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5. Practical 5

5.1. Practice Lab Assignment

5.1.1. Stacked Plot

5.2. Lab Assignment

5.2.1. Titanic Dataset

5.2.2. Histogram of passenger information ...

5.2.3. Bar plot of survival rate of passengers

5.2.4. Bar Plot for Survival by Gender

5.2.5. Bar Plot for Survival by Pclass

5.2.6. Bar Plot for Survival by Embarked

5.2.7. Box plot for Age Distribution

5.2.8. Box Plot for Age by Survived

5.2.9. Box Plot for Fare by Pclass

5.2.10. Scatter Plot for Age vs. Fare

5.2.11. Scatter Plot for Age vs. Fare by Sur...

5.2.1. Titanic Dataset

03:20

Write a Python program to analyze and visualize data from the Titanic dataset based on the following instructions:

Dataset Information:

The dataset is stored in a CSV file named `titanic.csv` and has been loaded using the `pandas` library. It contains the following columns:

- `Pclass`: Passenger class (1 = First, 2 = Second, 3 = Third).
- `Gender`: Gender of the passenger (male/female).
- `Age`: Age of the passenger.
- `Survived`: Survival status (0 = Did not survive, 1 = Survived).
- `Fare`: Ticket fare paid by the passenger.

Visualization:

To represent these trends, you will create 5 visualizations using `Matplotlib`. The visualizations should be arranged in a 3x2 grid (3 rows and 2 columns).

Visualization Details:

Write the code to create a series of visualizations as follows:

Bar Plot (Pclass Distribution):

- Create a bar plot to show the distribution of passengers across the different passenger classes (`Pclass`).
- Use the color `skyblue` for the bars.
- Title the plot as **"Passenger Class Distribution"**.
- Label the x-axis as **"Pclass"** and the y-axis as **"Count"**.

Pie Chart (Gender Distribution)

- Create a pie chart to display the distribution of male and female passengers.
- Use `lightblue` for males and `lightcoral` for females.
- Include percentages on the slices (use `autopct='%1.1f%%'`).
- Title the plot as **"Gender Distribution"**.

Histogram (Age Distribution):

- Create a histogram to visualize the distribution of passengers' ages.
- Use `lightgreen` for the bars with black edges (`edgecolor = 'black'`).

titanicDat...

Submit

1

import pandas as pd

2

import matplotlib.pyplot as plt

3

4

Load the Titanic dataset from the CSV file

5

df = pd.read_csv('titanic.csv')

6

7

Set up the figure for 5 subplots

8

fig, axes = plt.subplots(3, 2, figsize=(12, 12))

9

10

plt.hist(data['Age'],bins=30, edgecolor='k')

11

plt.title('Age Distribution')

12

plt.xlabel('Age')

13

plt.ylabel('Frequency')

14

plt.show()

15

import pandas as pd

16

import matplotlib.pyplot as plt

17

18

Load the Titanic dataset from the CSV file

19

df = pd.read_csv('titanic.csv')

20

Average time

0.520 s

Maximum time

0.520 s

1 out of 1 shown test case(s) passed

Test case 1

520 ms

Debug

Expected output

Actual output

Terminal

Test cases

30°C

Sunny

Search

ENG

IN

09:51

07-05-2026

Essentials of Data Science Laboratory - 2304102L - 2026

Search course ctrl + k

- 4.2.5. Titanic Dataset Analysis and Data Cl...
- 4.2.6. Titanic Dataset Analysis and Data Cl...
- 4.2.7. Titanic Dataset Analysis and Data Cl...
- 4.2.8. Titanic Dataset Analysis and Data Cl...

5. Practical 5

- 5.1. Practice Lab Assignment
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 - 5.2.1. Titanic Dataset
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 - 5.2.3. Bar plot of survival rate of passengers
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 - 5.2.7. Box plot for Age Distribution
 - 5.2.8. Box Plot for Age by Survived
 - 5.2.9. Box Plot for Fare by Pclass
 - 5.2.10. Scatter Plot for Age vs. Fare
 - 5.2.11. Scatter Plot for Age vs. Fare by Sur...

5.2.2. Histogram of passenger information of Titanic

Write a Python code to plot a histogram for the distribution of the 'Age' column from the Titanic dataset. The histogram should display the frequency of different age ranges with the following specifications:

1. Use **30 bins** for the histogram.
2. Set the **edge color** of the bars to **black (k)**.
3. Label the x-axis as **'Age'** and the y-axis as **'Frequency'**.
4. Add the title **"Age Distribution"** to the histogram.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4503,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

Note: Refer to the visible test case for better reference.

Sample Test Cases

Histogram...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Histogram
17 plt.hist(data['Age'], bins=30, edgecolor='k')
18 plt.title('Age Distribution')
19 plt.xlabel('Age')
20 plt.ylabel('Frequency')
```

Average time: 0.156 s, Maximum time: 0.156 s, 1 out of 1 shown test case(s) passed

Test case 1 (156 ms) Debug

Expected output: Age Distribution histogram

Actual output: Age Distribution histogram

Terminal Test cases

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4.2.5. Titanic Dataset Analysis and Data Cl...

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4.2.7. Titanic Dataset Analysis and Data Cl...

4.2.8. Titanic Dataset Analysis and Data Cl...

5. Practical 5

5.1. Practice Lab Assignment

5.1.1. Stacked Plot

5.2. Lab Assignment

5.2.1. Titanic Dataset

5.2.2. Histogram of passenger information ...

5.2.3. Bar plot of survival rate of passengers

5.2.4. Bar Plot for Survival by Gender

5.2.5. Bar Plot for Survival by Pclass

5.2.6. Bar Plot for Survival by Embarked

5.2.7. Box plot for Age Distribution

5.2.8. Box Plot for Age by Survived

5.2.9. Box Plot for Fare by Pclass

5.2.10. Scatter Plot for Age vs. Fare

5.2.11. Scatter Plot for Age vs. Fare by Sur...

5.2.3. Bar plot of survival rate of passengers

00:40

Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset. The chart should display the following specifications:

1. Use the 'Survived' column to show the count of survivors (0 = Did not survive, 1 = Survived).

2. Set the chart type to 'bar'.

3. Add the title "Survival Count" to the chart.

4. Label the x-axis as 'Survived' and the y-axis as 'Count'.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4503,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

BarPlotOf...

Submit

1 # import pandas as pd

2 # import matplotlib.pyplot as plt

3

4 # Load the Titanic dataset

5 # data = pd.read_csv('Titanic-Dataset.csv')

6

7 # Data Cleaning

8 # data['Age'].fillna(data['Age'].median(), inplace=True)

9 # data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

10 # data.drop('Cabin', axis=1, inplace=True)

11

12 # Convert categorical features to numeric

13 # data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

14 # data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

15

16 # Write your code here for Bar-Plot for Survival Rate

17

18 import pandas as pd

19 import matplotlib.pyplot as plt

20

Average time

0.187 s

187.00 ms

Maximum time

0.187 s

187.00 ms

1 out of 1 shown test case(s) passed

Test case 1 187 ms

Debug

Expected output

Actual output

Survival Count

Survival Count

Terminal

Test cases

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5. Practical 5

5.1. Practice Lab Assignment
5.1.1. Stacked Plot

5.2. Lab Assignment
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5.2.9. Box Plot for Fare by Pclass
5.2.10. Scatter Plot for Age vs. Fare
5.2.11. Scatter Plot for Age vs. Fare by Sur...

5.2.4. Bar Plot for Survival by Gender

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by gender, in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the 'Sex' column, then use the `value_counts()` function to count the occurrences of survivors (0 = Did not survive, 1 = Survived) for each gender.
2. Use a **stacked bar chart** to display the survival counts.
3. Add the title **"Survival by Gender"** to the chart.
4. Label the x-axis as **'Gender'** and the y-axis as **'Count'**.
5. The legend should indicate **'Not Survived'** and **'Survived'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

BarPlotOf...

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 survival_counts=data.groupby('Sex')['Survived'].value_counts().unstack()
17 survival_counts.plot(kind='bar',stacked=True)
18 plt.title("Survival by Gender")
19 plt.xlabel('Gender')
20 plt.ylabel('Count')

Average time 0.205 s
Maximum time 0.205 s
1 out of 1 shown test case(s) passed

Test case 1 205 ms
Expected output
Actual output

Survival by Gender
Survival by Gender

Terminal Test cases

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4.2.5. Titanic Dataset Analysis and Data Cl...

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5. Practical 5

5.1. Practice Lab Assignment

5.1.1. Stacked Plot

5.2. Lab Assignment

5.2.1. Titanic Dataset

5.2.2. Histogram of passenger information ...

5.2.3. Bar plot of survival rate of passengers

5.2.4. Bar Plot for Survival by Gender

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5.2.6. Bar Plot for Survival by Embarked

5.2.7. Box plot for Age Distribution

5.2.8. Box Plot for Age by Survived

5.2.9. Box Plot for Fare by Pclass

5.2.10. Scatter Plot for Age vs. Fare

5.2.11. Scatter Plot for Age vs. Fare by Sur...

5.2.5. Bar Plot for Survival by Pclass

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by passenger class (Pclass), in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the **Pclass** column and count the number of survivors (0 = Did not survive, 1 = Survived) for each class using **value_counts()**.

2. Use a **stacked bar chart** to display the survival counts.

3. Add the title **"Survival by Pclass"** to the chart.

4. Label the x-axis as **'Pclass'** and the y-axis as **'Count'**.

5. The legend should indicate **'Not Survived'** and **'Survived'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note:

Refer to the visible test case for better reference.

Ensure you use the **groupby()** function with **value_counts()** to count the survivors and non

BarPlotOf...

import pandas as pd

import matplotlib.pyplot as plt

Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')

Data Cleaning

data['Age'].fillna(data['Age'].median(), inplace=True)

data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

data.drop('Cabin', axis=1, inplace=True)

Convert categorical features to numeric

data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

Write your code here for Bar Plot for Survival by Pclass

survival_counts=data.groupby('Pclass')

['Survived'].value_counts().unstack()

survival_counts.plot(kind='bar',stacked=True)

plt.title('Survival by Pclass')

Average time

0.219 s

219.00 ms

Maximum time

0.219 s

219.00 ms

1 out of 1 shown test case(s) passed

Test case 1 219 ms

Debug

Expected output

Survival by Pclass

Actual output

Survival by Pclass

Terminal

Test cases

5.2.6. Bar Plot for Survival by Embarked

Write a Python code to plot a stacked bar chart showing the survival count for passengers based on their embarkation location in the Titanic dataset. The chart should display the following specifications:

- Use the **Embarked** column to determine the embarkation location. After converting this column into dummy variables (using **pd.get_dummies()**), plot the survival count based on the **Embarked_Q** column (representing passengers who embarked from Queenstown) in relation to survival.
- Set the chart type to 'bar' and make it stacked.
- Add the title "**Survival by Embarked**" to the chart.
- Label the x-axis as "**Embarked**" and the y-axis as "**Count**".
- Include a legend to distinguish between survivors and non-survivors (label the legend as '**Survived**' and '**Not Survived**').

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17403,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

BarPlotOf...

```
1 ## import pandas as pd
2 ## import matplotlib.pyplot as plt
3
4 ## ## Load the Titanic dataset
5 ## data = pd.read_csv('Titanic-Dataset.csv')
6
7 ## ## Data Cleaning
8 ## data['Age'].fillna(data['Age'].median(), inplace=True)
9 ## data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 ## data.drop('Cabin', axis=1, inplace=True)
11
12 ## ## Convert categorical features to numeric
13 ## data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 ## data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 ## ## Write your code here for Bar Plot for Survival by Embarked
17
18
19 # import pandas as pd
20 # import matplotlib.pyplot as plt
```

Average time 0.211 s 211.00 ms Maximum time 0.211 s 211.00 ms 1 out of 1 shown test case(s) passed

Test case 1 211 ms Debug

Expected output

Actual output

Essentials of Data Science

Laboratory - 2304102L - 2026

Search course

4.2.5. Titanic Dataset Analysis and Data Cl...

4.2.6. Titanic Dataset Analysis and Data Cl...

4.2.7. Titanic Dataset Analysis and Data Cl...

4.2.8. Titanic Dataset Analysis and Data Cl...

5. Practical 5

5.1. Practice Lab Assignment

5.1.1. Stacked Plot

5.2. Lab Assignment

5.2.1. Titanic Dataset

5.2.2. Histogram of passenger information ...

5.2.3. Bar plot of survival rate of passengers

5.2.4. Bar Plot for Survival by Gender

5.2.5. Bar Plot for Survival by Pclass

5.2.6. Bar Plot for Survival by Embarked

5.2.7. Box plot for Age Distribution

5.2.8. Box Plot for Age by Survived

5.2.9. Box Plot for Fare by Pclass

5.2.10. Scatter Plot for Age vs. Fare

5.2.11. Scatter Plot for Age vs. Fare by Sur...

5.2.7. Box plot for Age Distribution

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset across different passenger classes. The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot

2. Set the title of the plot to **"Age by Pclass"**.

3. Remove the default subtitle with **plt.suptitle("")**.

4. Label the x-axis as **'Pclass'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,0,0,330877,8.4503,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

BoxPlotF...

1 # import pandas as pd

2 # import matplotlib.pyplot as plt

3

4

5 # Load the Titanic dataset

6 # data = pd.read_csv('Titanic-Dataset.csv')

7

8 # Data Cleaning

9 # data['Age'].fillna(data['Age'].median(), inplace=True)

10 # data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

11 # data.drop('Cabin', axis=1, inplace=True)

12

13 # Convert categorical features to numeric

14 # data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

15 # data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

16

17 # Write your code here for Box Plot for Age by Pclass

18

19 import pandas as pd

20 import matplotlib.pyplot as plt

Average time

0.177 s

177.00 ms

Maximum time

0.177 s

177.00 ms

1 out of 1 shown test case(s) passed

Test case 1 177 ms

Debug

Expected output

Actual output

Age by Pclass

Age by Pclass

Terminal

Test cases

Essentials of Data Science

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Search course

ctrl + k

4.2.5. Titanic Dataset Analysis and Data Cl...

4.2.6. Titanic Dataset Analysis and Data Cl...

4.2.7. Titanic Dataset Analysis and Data Cl...

4.2.8. Titanic Dataset Analysis and Data Cl...

5. Practical 5

5.1. Practice Lab Assignment

5.1.1. Stacked Plot

5.2. Lab Assignment

5.2.1. Titanic Dataset

5.2.2. Histogram of passenger information ...

5.2.3. Bar plot of survival rate of passengers

5.2.4. Bar Plot for Survival by Gender

5.2.5. Bar Plot for Survival by Pclass

5.2.6. Bar Plot for Survival by Embarked

5.2.7. Box plot for Age Distribution

5.2.8. Box Plot for Age by Survived

5.2.9. Box Plot for Fare by Pclass

5.2.10. Scatter Plot for Age vs. Fare

5.2.11. Scatter Plot for Age vs. Fare by Sur...

5.2.8. Box Plot for Age by Survived

07:47

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:

1. Use the **Survived** column to group the data for the boxplot (0 = Did not survive, 1 = Survived).

2. Set the title of the plot to **"Age by Survival"**.

3. Remove the default subtitle with **plt.suptitle("")**.

4. Label the x-axis as **'Survived'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

BoxPlotF...

Submit

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Box Plot for Age by Survived
17
18 data.boxplot(column='Age', by='Survived')
19 plt.title('Age by Survival')
20 plt.suptitle('')

Average time 0.200 s
Maximum time 0.200 s
1 out of 1 shown test case(s) passed

Test case 1 200 ms Debug

Expected output

Actual output

Terminal Test cases

Essentials of Data Science

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Search course

ctrl + k

5. Practical 5

5.1. Practice Lab Assignment

5.2. Lab Assignment

5.2.1. Titanic Dataset

5.2.2. Histogram of passenger information ...

5.2.3. Bar plot of survival rate of passengers

5.2.4. Bar Plot for Survival by Gender

5.2.5. Bar Plot for Survival by Pclass

5.2.6. Bar Plot for Survival by Embarked

5.2.7. Box plot for Age Distribution

5.2.8. Box Plot for Age by Survived

5.2.9. Box Plot for Fare by Pclass

5.2.10. Scatter Plot for Age vs. Fare

5.2.11. Scatter Plot for Age vs. Fare by Sur...

5.2.9. Box Plot for Fare by Pclass

08:05

Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot

2. Set the title of the plot to **"Fare by Pclass"**.

3. Remove the default subtitle with **plt.suptitle("")**.

4. Label the x-axis as **'Pclass'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4503,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

BoxPlotF...

Submit

1import pandas as pd

2import matplotlib.pyplot as plt

3

4# Load the Titanic dataset

5data = pd.read_csv('Titanic-Dataset.csv')

6

7# Data Cleaning

8data['Age'].fillna(data['Age'].median(), inplace=True)

9data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

10data.drop('Cabin', axis=1, inplace=True)

11

12# Convert categorical features to numeric

13data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

14data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

15

16# Write your code here for Box Plot for Fare by Pclass

17data.boxplot(column='Fare',by='Pclass')

18plt.title('Fare by Pclass')

19plt.suptitle('')

20plt.xlabel('Pclass')

21

Average time0.200 s

Maximum time0.200 s

1 out of 1 shown test case(s) passed

Test case 1200 ms

Debug

Expected output

Actual output

Terminal

Test cases

Essentials of Data Science
Laboratory - 2304102L - 2026

Search coursectrl + k

4.2.5. Titanic Dataset Analysis and Data Cl...

4.2.6. Titanic Dataset Analysis and Data Cl...

4.2.7. Titanic Dataset Analysis and Data Cl...

4.2.8. Titanic Dataset Analysis and Data Cl...

5. Practical 5

5.1. Practice Lab Assignment

5.1.1. Stacked Plot

5.2. Lab Assignment

5.2.1. Titanic Dataset

5.2.2. Histogram of passenger information ...

5.2.3. Bar plot of survival rate of passengers

5.2.4. Bar Plot for Survival by Gender

5.2.5. Bar Plot for Survival by Pclass

5.2.6. Bar Plot for Survival by Embarked

5.2.7. Box plot for Age Distribution

5.2.8. Box Plot for Age by Survived

5.2.9. Box Plot for Fare by Pclass

5.2.10. Scatter Plot for Age vs. Fare

5.2.11. Scatter Plot for Age vs. Fare by Sur...

5.2.10. Scatter Plot for Age vs. Fare08:01

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset. The scatter plot should display the following specifications:
1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Set the title of the plot to "**Age vs. Fare**".
3. Label the x-axis as "**Age**" and the y-axis as "**Fare**".

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Bround, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

AgeFare\$...

```
1 # import pandas as pd
2 # import matplotlib.pyplot as plt
3
4 ## Load the Titanic dataset
5 # data = pd.read_csv('Titanic-Dataset.csv')
6
7 ## Data Cleaning
8 # data['Age'].fillna(data['Age'].median(), inplace=True)
9 # data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 # data.drop('Cabin', axis=1, inplace=True)
11
12 ## Convert categorical features to numeric
13 # data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 # data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 ## Write your code here for Box Plot for Fare by Pclass
17
18
19 import pandas as pd
20 import matplotlib.pyplot as plt
```

Average time0.149 s149.00 msMaximum time0.149 s149.00 ms1 out of 1 shown test case(s) passed

Test case 1140 msDebug

Expected outputActual output

TerminalTest cases

Essentials of Data Science Laboratory - 2304102L - 2026

Search course ctrl + k

- 4.2.5. Titanic Dataset Analysis and Data Cl...
- 4.2.6. Titanic Dataset Analysis and Data Cl...
- 4.2.7. Titanic Dataset Analysis and Data Cl...
- 4.2.8. Titanic Dataset Analysis and Data Cl...

5. Practical 5

- 5.1. Practice Lab Assignment
 - 5.1.1. Stacked Plot
- 5.2. Lab Assignment
 - 5.2.1. Titanic Dataset
 - 5.2.2. Histogram of passenger information ...
 - 5.2.3. Bar plot of survival rate of passengers
 - 5.2.4. Bar Plot for Survival by Gender
 - 5.2.5. Bar Plot for Survival by Pclass
 - 5.2.6. Bar Plot for Survival by Embarked
 - 5.2.7. Box plot for Age Distribution
 - 5.2.8. Box Plot for Age by Survived
 - 5.2.9. Box Plot for Fare by Pclass
 - 5.2.10. Scatter Plot for Age vs. Fare
 - 5.2.11. Scatter Plot for Age vs. Fare by Sur...

5.2.11. Scatter Plot for Age vs. Fare by Survived

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Color the points based on the **Survived** column: **Red** for passengers who did not survive (**Survived = 0**). **Blue** for passengers who survived (**Survived = 1**).
3. Set the title of the plot to **"Age vs. Fare by Survival"**.
4. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

Note: Refer to the visible test case for better reference.

Sample Test Cases

AgeFareS...

```
## Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

## Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

## Write your code here for Scatter Plot for Age vs. Fare by Survived

import pandas as pd
import matplotlib.pyplot as plt

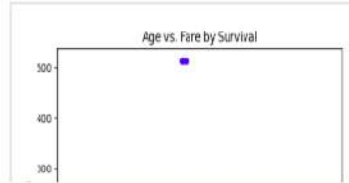
# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
```

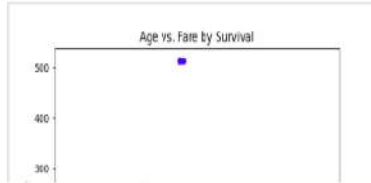
Average time: 0.161 s, 161.00 ms | Maximum time: 0.161 s, 161.00 ms | 1 out of 1 shown test case(s) passed

Test case 1 (161 ms) [Debug]

Expected output



Actual output



Terminal Test cases